

YaRN Multilingual Training Report

Full Training Run — SLURM Job 18536300

OpenEuroLLM Long-Context Pipeline · LUMI Supercomputer (CSC) · 2026-05-10

1. Job Summary

Job ID	18536300
Cluster	LUMI (CSC), Finland
Partition	standard-g (AMD Instinct MI250X GPUs)
Start time	2026-05-10 08:51 UTC
End time	2026-05-10 17:44 UTC
Wall time	~9 hours
Nodes	32
GPUs	256 (8 per node)
Account	project_462000963

2. Model & Training Configuration

Base model	OpenEuroLLM 9B (LlamaForCausalLM)
Architecture	32 layers · hidden 4096 · 32 heads · FFN 14336 · vocab 262 144
Context extension	YaRN · factor=16.0 · 2 048 → 32 768 tokens
Loaded checkpoint	/flash/project_462000963/jouni/checkpoints/oellm-9b-80-20-TP-2-PP-4
Parallelism	TP=2, PP=4, CP=4, Sequence Parallel, Distributed Optimizer
Sequence length	32 768 tokens
Global batch size	128
Micro batch size	1
Total iterations	1 000
Total tokens seen	~4.19 billion (128 × 32 768 × 1 000)
Optimizer	Adam $\beta_1=0.9$ $\beta_2=0.95$ $\epsilon=1e-8$
Learning rate	1e-5 peak · 1e-7 min · WSD schedule
Warmup / Cooldown	1/10 warmup · 1/5 WSD cooldown
Weight decay	0.05
Gradient clipping	1.0
Precision	BF16
Activation recompute	Yes (--recompute-activations)

3. Training Data

Pre-tokenized Megatron bin/idx files from HuggingFace dataset birgermoell/oellm-longctx-tokenized-streamed-all-v2, downloaded and merged on LUMI by download_tokenized.sbatch (job 18504569).

Languages	8: Bulgarian (bg), Czech (cs), Danish (da), Estonian (et), Finnish (fi), French (fr), Croatian (hr), Dutch (nl)
Tiers	16k_plus ($\geq 16\,384$ tokens) · 4_16k (4 096–16 383) · under4k ($< 4\,096$)
Merged files	24 (8 languages × 3 tiers)
Total size on disk	87 GB
Estimated tokens	~35 billion
DATA_PATH entries	24 (uniform per-language weighting per tier)
Tier weights	16k_plus: 0.50 · 4_16k: 0.30 · under4k: 0.20
Per-language weight	0.0625 (16k_plus) · 0.0375 (4_16k) · 0.025 (under4k)
Tokenizer	openeurollm/tokenizer-256k (vocab size 262 144)
Data location	/flash/project_462000963/bmoell/data_tokenized_hf_multilingual/

4. Training Loss Curve

999 iterations logged in total. Table shows every 100th iteration plus first and last. Throughput is measured on rank 255 (last pipeline stage).

Iter	LM Loss	Grad Norm	TFLOP/GPU	Notes
2	12.2211	73.530	16.1	warmup — slow due to dataset index build
100	7.2543	9.115	40.5	loss already −5 nats from start
200	5.3297	2.107	40.6	
300	4.8971	1.962	40.6	
400	4.4716	1.613	40.6	
500	4.0961	1.623	40.5	midpoint checkpoint saved
600	3.9516	1.310	40.5	
700	3.8473	0.904	40.5	
800	3.6850	1.098	40.5	
900	3.5562	0.639	40.5	LR cooldown (WSD) begins
999	3.6024	0.331	40.5	
1000	3.6643	0.331	40.5	final checkpoint saved

Final validation loss: 3.5679 | Validation PPL: 35.44

Final test loss: 3.4302 | Test PPL: 30.88

5. Analysis

The model began training at loss 12.22 — slightly below the random baseline of $\ln(262\,144) \approx 12.48$. This elevated starting loss is expected: the base checkpoint was pre-trained at 2 048-token context, and YaRN's modified rotary position embeddings produce unfamiliar attention patterns at 32K context on the first forward pass.

By iteration 100 the loss had already dropped to 7.25 — a fall of 5 nats in under an hour. This rapid early descent reflects the model quickly learning the structure of YaRN-scaled positions. Gradient norms peak at 73 in the first iterations then stabilise to 1–2 by iter 200, confirming smooth convergence with no gradient explosions.

The final training loss of 3.66 and validation PPL of 35.4 indicate the model has genuinely adapted to multilingual long-context data. The test PPL of 30.9 being lower than validation is consistent with the test split containing slightly longer and cleaner documents concentrated in the 16k_plus tier.

Throughput was rock-solid at 40.5–40.6 TFLOP/s/GPU across all 256 GPUs for the entire 9-hour run (~513 tok/s/GPU). The only slow iteration was iter 2 (16.1 TFLOP/s) due to dataset index building on rank 0; all subsequent iterations ran at ~32 seconds.

6. Outputs

Checkpoint iter 500	/flash/project_462000963/bmoell/yarn-multilingual/checkpoints/iter_0000500/
Checkpoint iter 1000	/flash/project_462000963/bmoell/yarn-multilingual/checkpoints/iter_0001000/
HuggingFace model	birgermoell/oellm-9b-yarn-multilingual-32k (pytorch_model.bin, 17 GB)
Tensorboard logs	/flash/project_462000963/bmoell/yarn-multilingual/tensorboard/
SLURM stdout	/scratch/project_462000963/bmoell/yarn-multilingual-18536300.out

7. Next Steps

1. Evaluate — OneRuler-OELLM

Run NIAH smoke eval (eval_oneruler_smoke.sbatch) at 4K / 16K / 32K context on fr, fi, cs to confirm the context window works on downstream retrieval tasks. Follow with full 8-language eval if results are positive.

2. Scale data — 27 remaining languages

Run tokenize_tiers.sbatch for ro, uk, sr, hu, pl and other FinePDFs-Edu languages not in the

pre-tokenized HF dataset. Retrain with a broader 35-language mix.

3. LongRoPE search

Run `longrope_search_tokenize.sbatch` + `longrope_search.sbatch` for multilingual RoPE factors optimised on the target distribution. If eval shows improvement over YaRN, retrain with `longrope_multilingual.sbatch`.

4. Publish

Upload evaluation results and model card to `birgermoell/oellm-9b-yarn-multilingual-32k` on HuggingFace. Contribute ROCm/MI250X Megatron fixes upstream to OpenEuroLLM/NVIDIA-Megatron-LM.